

EXPERIEMNT 20

Prolog Program – Planet Database

Aim

To write and execute a Prolog program to create a database containing information about planets and retrieve the planet details using Prolog queries.

Algorithm

1. Start GNU Prolog.
2. Define a predicate planet(Name, Type, Distance).
3. Store the planet name, type, and distance from the Sun as facts.
4. Load the program into GNU Prolog.
5. Enter a planet name as the input query.
6. Prolog searches the database for the matching planet.
7. Retrieve and display the corresponding planet information.
8. Display the result.
9. Stop.

Program

```
% Planet Database
```

```
planet('Mercury', 'Terrestrial', '57.9 million km').
```

```
planet('Venus', 'Terrestrial', '108.2 million km').
```

```
planet('Earth', 'Terrestrial', '149.6 million km').
```

```
planet('Mars', 'Terrestrial', '227.9 million km').
```

```
planet('Jupiter', 'Gas Giant', '778.5 million km').
```

```
planet('Saturn', 'Gas Giant', '1.43 billion km').
```

```
planet('Uranus', 'Ice Giant', '2.87 billion km').
```

```
planet('Neptune', 'Ice Giant', '4.50 billion km').
```

Input

```
planet('Earth', Type, Distance).
```

Output:

```
(16 ms) yes
| ?- planet('Earth', Type, Distance).

Distance = '149.6 million km'
Type = 'Terrestrial'

yes
| ?- |
```

Result

Thus, the Prolog program for the Planet Database was successfully implemented and executed, and the required planet information was retrieved successfully.